

Fig. 9.17 Separating noise contributed by the source resistance and circuit noise.

Note that the input-referred noise $v_{in(rms)}$ in (9.63) is not obtained by integrating the input-referred noise spectrum over some bandwidth. The rms input-referred noise divides the rms output noise by the mid-band gain only so that it can be used to determine SNR assuming the input signal is confined to the circuit's mid-band region.

In radio-frequency applications, the signal input is narrowband and filters can be used to eliminate all out-of-band noise. In this case, a measure of noise spectral density at a particular frequency, also called the *spot noise*, is useful. A common noise metric in this case is the *noise factor*, $F(f)$, given by the ratio of the total noise to the noise that would arise if only the source resistance were noisy, at a particular frequency.

$$F(f) = \frac{v_{no}^2(f)}{4kTR_s|A(f)|^2} \quad (9.66)$$

This quantity may be related to the input referred noise of the amplifier alone, $v_{ai}^2(f)$.

$$F(f) = \frac{4kTR_s + v_{ai}^2(f)}{4kTR_s} = 1 + \frac{v_{ai}^2(f)}{4kTR_s} \quad (9.67)$$

Expressed in decibels, the noise factor becomes the noise figure of the circuit,

$$NF(f) = 10\log_{10}[F(f)] \text{ dB} \quad (9.68)$$

Key Point: The noise performance of a circuit at a particular frequency is usually reported as its input-referred noise spectrum, equal to the circuit's output noise spectrum divided by its magnitude response, or equivalently by its noise figure.

9.4 NOISE ANALYSIS EXAMPLES

In this section, a variety of circuits are analyzed from a noise perspective. Although some useful design techniques for reducing noise are presented, the main purpose of this section is to give the reader some experience in analyzing circuits from a noise perspective.

9.4.1 Opamp Example

Consider an inverting amplifier and its equivalent noise model, as shown in Fig. 9.18. Here, $V_n(f)$, $I_{n-}(f)$, and $I_{n+}(f)$ represent the opamp's equivalent input noise, and the remaining noise sources are resistor thermal noise sources. Note that current noise sources are used in the models for R_1 and R_f , whereas a voltage noise source is used for R_2 . As we will see, these choices of noise sources simplify the circuit analysis.

First, using superposition and assuming all noise sources are uncorrelated, consider the output noise voltage, $V_{no1}^2(f)$, due only to I_{n1} , I_{n1} , and I_{n-} . These three noise currents add together, and their total current sum is fed into the parallel combination of C_f and R_f . Note that no current passes through R_1 since the voltage across it is zero due to the virtual ground at the negative opamp terminal (assuming a high-gain opamp). Thus, the output noise mean-squared value due to these three noise currents is given by

$$V_{no1}^2(f) = [I_{n1}^2(f) + I_{n1}^2(f) + I_{n-}^2(f)] \left| \frac{R_f}{1 + j2\pi f R_f C_f} \right|^2 \quad (9.69)$$

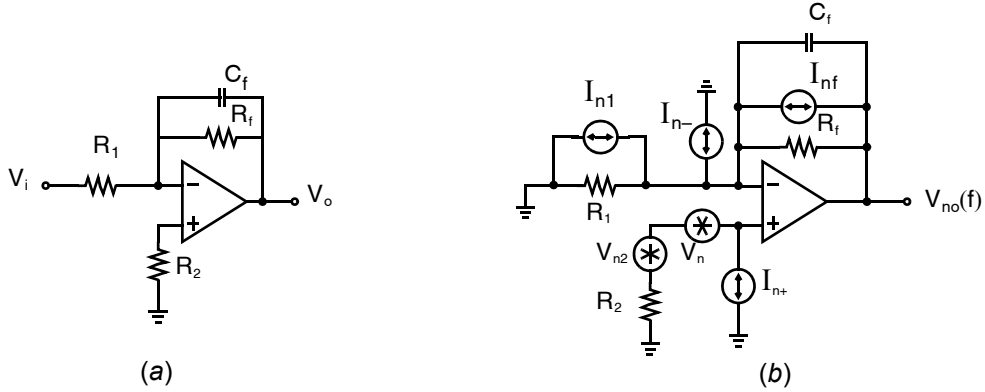


Fig. 9.18 (a) Low-pass filter, and (b) equivalent noise model.

This equation indicates that this part of the output noise value equals the sum of the noise currents multiplied by R_f^2 . This noise portion is then shaped by a low-pass filter with a 3-dB frequency equal to $f_0 = 1/(2\pi R_f C_f)$. Following an analysis similar to that in Section 9.3.6, the total integrated mean squared output noise voltage due to R_f alone may be shown to be simply kT/C_f . Including the noise of R_1 increases the result to $(1 + R_f/R_1)kT/C_f$ and including opamp noise increases it further. Hence, kT/C_f is a lower limit on the noise of opamps with capacitive feedback.

Using superposition further, the output mean-squared value, $V_{no2}^2(f)$, due to the three noise sources at the positive opamp terminal can be found as follows: By converting I_{n+} to a voltage source (by multiplying it by R_2), we see that the three noise voltages are summed and are applied to the positive opamp terminal. Since the gain from the positive opamp terminal to the output signal is easily found, the output noise mean-squared value due to these three noise sources is given by

$$V_{no2}^2(f) = [I_{n+}^2(f)R_2^2 + V_{n2}^2(f) + V_n^2(f)] \left| 1 + \frac{R_f/R_1}{1 + j2\pi f C_f R_f} \right|^2 \quad (9.70)$$

This equation indicates that this part of the output noise mean-squared value equals the sum of the noise voltages, and this noise portion is then shaped by the shown transfer function. For this transfer function, if $R_f \ll R_1$, then its gain approximately equals unity for all frequencies. Thus, for an ideal opamp, the noise would exist up to infinite frequency, resulting in an infinite amount of mean-squared volts. However, for practical circuits, the gain drops off above the unity-gain frequency of the opamp, and thus the noise is effectively low-pass filtered. In the case where $R_f \gg R_1$, the low-frequency gain is roughly R_f/R_1 , and its 3-dB frequency is the same as in the case of the noise sources at the negative opamp terminal (i.e., $f_0 = 1/(2\pi R_f C_f)$). However, in this case, the gain only decreases to unity and then remains at that level. Treating the Bode plot as exact and noting that, above f_0 , the gain drops off at -20 dB/decade, this transfer function reaches unity around $f_1 = (R_f/R_1)f_0$. Thus, one should also include the opamp's positive input noise (with a gain of one) integrated over the region between f_1 and the unity-gain frequency of the opamp.

Finally, the total output noise mean-squared value is simply the sum

$$V_{no}^2(f) = V_{no1}^2(f) + V_{no2}^2(f) \quad (9.71)$$

or, if rms values are found,

$$V_{no(rms)}^2 = V_{no1(rms)}^2 + V_{no2(rms)}^2 \quad (9.72)$$

EXAMPLE 9.10

Estimate the total output noise rms value for a 10-kHz low-pass filter, as shown in Fig. 9.18, when $C_f = 160$ pF, $R_f = 100$ k, $R_1 = 10$ k, and $R_2 = 9.1$ k. Also, find the SNR for an input signal equal to 100 mV rms. Assume that the noise voltage of the opamp is given by $V_n(f) = 20$ nV/ $\sqrt{\text{Hz}}$, both its noise currents are $I_n(f) = 0.6$ pA/ $\sqrt{\text{Hz}}$, and that its unity-gain frequency equals 5 MHz.

Solution

Assuming the device is at room temperature, the resistor noise sources are equal to

$$I_{nf} = 0.406 \text{ pA}/\sqrt{\text{Hz}} \quad (9.73)$$

$$I_{n1} = 1.28 \text{ pA}/\sqrt{\text{Hz}} \quad (9.74)$$

$$V_{n2} = 12.2 \text{ nV}/\sqrt{\text{Hz}} \quad (9.75)$$

The low-frequency value of $V_{no1}^2(f)$ is found by letting $f = 0$ in (9.69).

$$\begin{aligned} V_{no1}^2(0) &= [I_{n1}^2(0) + I_{nf}^2(0) + I_{n-}(0)]R_f^2 \\ &= (0.406^2 + 1.28^2 + 0.6^2)(1 \times 10^{-12})^2(100 \text{ k})^2 \\ &= (147 \text{ nV}/\sqrt{\text{Hz}})^2 \end{aligned} \quad (9.76)$$

Since (9.69) also indicates that this noise is low-pass filtered, the rms output noise value due to these three sources can be found using the concept of a noise equivalent bandwidth. Specifically, we multiply the spectral density by $(\pi f_0)/2$, where $f_0 = 1/(2\pi R_f C_f)$. Thus,

$$\begin{aligned} V_{no1(rms)}^2 &= (147 \text{ nV}/\sqrt{\text{Hz}})^2 \times \frac{1}{4(100 \text{ k}\Omega)(160 \text{ pF})} \\ &= (18.4 \text{ }\mu\text{V})^2 \end{aligned} \quad (9.77)$$

To estimate the output noise due to the sources at the positive opamp terminal, we find $V_{no2}^2(0)$ to be given by

$$\begin{aligned} V_{no2}^2(0) &= [I_{n+}^2(f)R_2^2 + V_{n2}^2(f) + V_n^2(f)](1 + R_f/R_1)^2 \\ &= (24.1 \text{ nV}/\sqrt{\text{Hz}})^2 \times 11^2 \\ &= (265 \text{ nV}/\sqrt{\text{Hz}})^2 \end{aligned} \quad (9.78)$$

This noise is also low-pass filtered at f_0 until $f_1 = (R_f/R_1)f_0$, where the noise gain reaches unity and it remains until $f_t = 5$ MHz (i.e., the unity-gain frequency of the opamp). Thus, breaking this noise into two portions, and using (9.40) to calculate the first portion, we have

$$\begin{aligned} V_{no2(rms)}^2 &= (265 \times 10^{-9})^2 \left(\frac{1}{4R_f C_f} \right) + (24.1 \times 10^{-9})^2 \left(\frac{\pi}{2} \right) (f_t - f_1) \\ &= (74.6 \text{ }\mu\text{V})^2 \end{aligned} \quad (9.79)$$

Thus, the total output noise is estimated to be

$$V_{no(rms)} = \sqrt{V_{no1(rms)}^2 + V_{no2(rms)}^2} = 77 \text{ }\mu\text{V rms} \quad (9.80)$$

It should be noted here that the major source of noise at low frequencies is due to the opamp's voltage noise, $V_n(f)$.

To obtain the SNR for this circuit with a 100-mV rms input level, one can find the output signal mean-squared value and relate that to the output noise value. Alternatively, one can find the equivalent input noise value

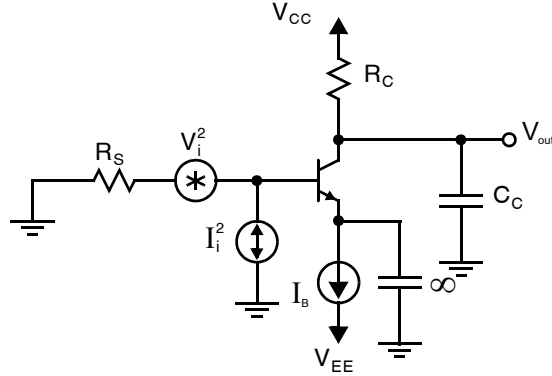


Fig. 9.19 A bipolar common-emitter amplifier.

by dividing the output noise value by the circuit's gain and then relate the input signal and noise mean-squared values. Taking the first approach results in an output signal level of 1 V rms, which gives an SNR of

$$\text{SNR} = 20 \log \left(\frac{1 \text{ V}}{77 \mu\text{V}} \right) = 82 \text{ dB} \quad (9.81)$$

Note here that using a lower-speed opamp would have reduced the total output noise, as would choosing an opamp with a lower noise voltage. Also note that R_2 contributes to the output noise through both its thermal noise and the noise current of the opamp's positive input. Since its only purpose is to improve the dc offset performance, R_2 should be eliminated in a low-noise circuit (assuming dc offset can be tolerated).

9.4.2 Bipolar Common-Emitter Example

In this example, we consider a bipolar common-emitter amplifier, as shown in Fig. 9.19. Here, we wish to find the optimum bias current to minimize the equivalent input noise of this amplifier (i.e., maximize the signal-to-noise ratio of the amplifier).

We should state that it is assumed here that the collector-current shot noise dominates in the input voltage noise source and the base-current shot noise dominates in the input current noise source. Therefore, the input voltage noise due to the transistor alone is given by

$$V_i^2(f) = 2kT/g_m \quad (9.82)$$

and the input current noise due to the transistor alone is given by

$$I_i^2(f) = 2qI_B \quad (9.83)$$

These assumptions are reasonable since usually $1/f$ noise is not important for bipolar transistors, and the collector shot noise is usually not a major component of the input current noise source for wideband examples. Also, although the base resistance has temporarily been ignored, it will be taken into account shortly by simply modifying the size of the source resistance.

The first step is to replace the input current noise source by an input voltage noise source. First, note that the gain from the base to the collector (ignoring the transistor output impedance, r_o) is equal to $g_m R_C$, and the impedance looking into the base is r_π . Therefore, the output noise due to $I_i^2(f)$ is given by

$$V_{oi}^2(f) = I_i^2(f) [(R_S \parallel r_\pi) g_m R_C]^2 \quad (9.84)$$